ENTERPRISE VIDEO TRANSCRIPTION PIPELINE

Dependency Failure Resolution Playbook

Root Cause: NumPy 2.0 / Setuptools 80+ / HF Hub 1.29 vs Pyannote 3.1.1 | Fix: Locked Stack 3.3.2

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Date	**September 3, 2026 | 14:35-14:47 IST | ~12 mins for 2 videos**
Audience	**Junior Engineer / New Joiner - Zero Prior Context (18-year shelf life)**
Status	**RESOLVED - pyannote ok + Transcripts Generated**
Tool used to debug the issue	Meta ai

1. Executive Summary

Problem: The enterprise video transcription pipeline, which converts meeting videos into timestamped transcripts with speaker labels using Whisper and Pyannote, failed to initialize. The system was built using a 2023 tutorial referencing pyannote.audio 3.1.1, but was executed in a 2026 Python environment where pip installs the latest package versions by default. This version drift triggered seven cascading import and compatibility errors, blocking transcription entirely.

Root Cause: Four major breaking releases occurred between 2023 and 2026. First, NumPy 2.0 removed np.NaN and np.Inf. Second, Setuptools 80+ removed pkg_resources. Third, Hugging Face Hub 1.0 broke <1.0 APIs. Fourth, Torchvision 0.20 introduced circular import when torch and torchvision versions mismatch. No requirements_locked.txt existed.

Solution: Locked entire stack to compatible era: pyannote.audio 3.3.2 fixes NumPy 2.0, torch 2.4.1 + torchvision 0.19.1 + torchaudio 2.4.1 as atomic unit, setuptools 69.0.3, huggingface_hub 0.35.3, matplotlib. Two code adapters handle breaking API changes.

Validation: python -c "from pyannote.audio import Pipeline; print('pyannote ok')" returned pyannote ok. Two videos transcribed in ~12 mins on CUDA.

2. What Is This Pipeline? (For Reader in 2044)

- **ffmpeg:** System binary that extracts audio from video. Must be in PATH. Converts mp4 to 16kHz mono WAV.
- **Whisper:** OpenAI AI model for speech-to-text. Takes WAV, returns text + timestamps. Medium size ~1.5GB.
- **Pyannote speaker-diarization-3.1:** AI model that determines who spoke when. Returns Annotation with itertracks() in 3.1.1, DiarizeOutput wrapper in 3.3.2.
- **Alignment Logic:** Overlaps Whisper segments with Pyannote turns to assign speaker to each sentence.
- **Output:** TXT file with [Speaker] [HH:MM:SS.mmm] text + Start/End Time + Duration in IST, self-describing for audit.

3. Pre-Requisites Checklist

- **Python 3.11:** Command: python --version | Expected: Python 3.11.x
- **ffmpeg:** Command: ffmpeg -version | Expected: ffmpeg version N-...
- **CUDA GPU:** Command: nvidia-smi | Expected: GPU list
- **HF Account + License:** Command: Create token at huggingface.co/settings/tokens + Accept at huggingface.co/pyannote/speaker-diarization-3.1 | Expected: Access Granted

4. Diagnostic Playbook - Color-Coded

Color Legend for Engagement and Clarity:

- **BLUE Bold Consolas 10.5pt = COMMAND TO RUN**
- **GREEN Italic Consolas 9.5pt = EXPECTED OUTPUT**
- **RED Bold Consolas 9.5pt = ACTUAL ERROR**
- **ORANGE Calibri = WHERE TO CHECK**
- **DARK GREEN Bold = HOW TO FIX**

STEP 1: Is Virtual Environment Active?

COMMAND TO RUN:

```
where python   (Windows)   OR   which python   (Linux/Mac)
```

EXPECTED OUTPUT (If OK):

```
D:\...\venv\Scripts\python.exe   (path MUST contain 'venv')
```

ACTUAL ERROR (If Broken):

```
C:\Python311\python.exe   (system Python, no venv)
```

WHERE TO CHECK: Check prompt shows (venv). If not, venv not active. Run: echo %VIRTUAL_ENV%

HOW TO FIX: .\venv\Scripts\Activate.ps1 (Windows) OR source venv/bin/activate (Linux/Mac)

WHY THIS HAPPENS & WHY MULTIPLE INSTALLS NEEDED: If venv not active, pip installs globally, breaking other projects. 40% of env issues are venv not active.

STEP 2: Is Pyannote Importable?

COMMAND TO RUN:

```
python -c "from pyannote.audio import Pipeline; print('pyannote ok')"
```

EXPECTED OUTPUT (If OK):

```
pyannote ok + harmless UserWarning about torchaudio backend - IGNORE warning
```

ACTUAL ERROR (If Broken):

```
ModuleNotFoundError: No module named 'pyannote'
```

WHERE TO CHECK: pip list | findstr pyannote -> Should show pyannote.audio 3.3.2

HOW TO FIX: pip install pyannote.audio==3.3.2

WHY THIS HAPPENS & WHY MULTIPLE INSTALLS NEEDED: Fastest health check. If fails, fix env first. Add as CI gate.

STEP 3: NumPy 2.0 Check - Silent Killer

COMMAND TO RUN:

```
python -c "import numpy; print(numpy.__version__)"
```

EXPECTED OUTPUT (If OK):

```
2.4.6 (NEW stack with 3.3.2) OR 1.26.4 (OLD stack with 3.1.1)
```

ACTUAL ERROR (If Broken):

```
AttributeError: `np.NaN` was removed in NumPy 2.0 release. Use `np.nan` instead. File
inference.py line 533 missing: float = np.NaN
```

WHERE TO CHECK: pyannote/audio/core/inference.py line 533 uses np.NaN. NumPy 2.0 June 2024 removed aliases.

HOW TO FIX: RECOMMENDED: pip install pyannote.audio==3.3.2 (fixes np.NaN->np.nan) | LEGACY: pip install numpy==1.26.4 BUT then must downgrade pyannote-core to 5.0.0

WHY THIS HAPPENS & WHY MULTIPLE INSTALLS NEEDED: NumPy 2.0 removed np.NaN, np.Inf. Library using them breaks. Upgrade library, not downgrade NumPy.

STEP 4: Setuptools pkg_resources Check

COMMAND TO RUN:

```
python -c "import pkg_resources; print('ok')"
```

EXPECTED OUTPUT (If OK):

```
ok
```

ACTUAL ERROR (If Broken):

```
ModuleNotFoundError: No module named 'pkg_resources' File
lightning_fabric/__init__.py line 41
```

WHERE TO CHECK: pip show setuptools -> If 80.x or 75.x, broken. Expected 69.0.3. Setuptools changelog: pkg_resources removed in 80+

HOW TO FIX: pip install setuptools==69.0.3 wheel==0.44.0 - 69 is last with pkg_resources

WHY THIS HAPPENS & WHY MULTIPLE INSTALLS NEEDED: lightning_fabric 2.2.5 imports pkg_resources deleted in setuptools 80+. Never upgrade setuptools on old projects.

STEP 5: Torch Trio Atomic Check

COMMAND TO RUN:

```
python -c "import torch, torchvision, torchaudio;
print(torch.__version__, torchvision.__version__)"
```

EXPECTED OUTPUT (If OK):

```
2.4.1 0.19.1 2.4.1 (must match from same CUDA build)
```

ACTUAL ERROR (If Broken):

```
AttributeError: partially initialized module 'torchvision' has no attribute
'extension' (circular import) File torchvision/_meta_registrations.py line 25
```

WHERE TO CHECK: torchvision/_meta_registrations.py imports extension at import time. Mismatch causes circular import.

HOW TO FIX: pip uninstall -y torch torchvision torchaudio | pip install torch==2.4.1 torchvision==0.19.1 torchaudio==2.4.1 --index-url https://download.pytorch.org/whl/cu121 - Install all 3 TOGETHER

WHY THIS HAPPENS & WHY MULTIPLE INSTALLS NEEDED: Torch trio must be upgraded atomically. Installing one alone upgrades to 2.5.1 but leaves others at 0.19.1 -> mismatch. This is why multiple uninstall/reinstall cycles were needed.

STEP 6: HF Hub Version Check

COMMAND TO RUN:

```
pip show huggingface_hub
```

EXPECTED OUTPUT (If OK):

```
Version: 0.35.3 (must be <1.0)
```

ACTUAL ERROR (If Broken):

```
ERROR: transformers requires huggingface-hub<1.0 but you have 1.29.0
```

WHERE TO CHECK: pip list | huggingface_hub 1.29.0 is latest 2026 but breaking. Many libs require <1.0

HOW TO FIX: pip install huggingface_hub==0.35.3 hf-xet==1.1.5 - 0.35.3 is last stable <1.0

WHY THIS HAPPENS & WHY MULTIPLE INSTALLS NEEDED: HF Hub 1.0 was breaking release. Old libs not migrated. Latest is not greatest.

STEP 7: Matplotlib Check

COMMAND TO RUN:

```
python -c "import matplotlib; print('ok')"
```

EXPECTED OUTPUT (If OK):

```
ok
```

ACTUAL ERROR (If Broken):

```
ModuleNotFoundError: No module named 'matplotlib' during Pipeline.from_pretrained
```

WHERE TO CHECK: Traceback shows pyannote pipeline loading fails at import matplotlib

HOW TO FIX: pip install matplotlib

WHY THIS HAPPENS & WHY MULTIPLE INSTALLS NEEDED: Pyannote uses matplotlib even if you don't plot. Required dependency but not declared hard.

STEP 8: HF Token & License Check

COMMAND TO RUN:

```
hf auth whoami
```

EXPECTED OUTPUT (If OK):

```
YourUsername + Browser shows model files, not Agree button
```

ACTUAL ERROR (If Broken):

```
401 Unauthorized / 403 Forbidden / Gated repo - You must be authenticated
```

WHERE TO CHECK: Browser: https://huggingface.co/pyannote/speaker-diarization-3.1 - Shows Agree button? If YES, not accepted.

HOW TO FIX: Browser: Click Agree on pyannote/speaker-diarization-3.1 and pyannote/segmentation-3.0 | Terminal: hf auth login paste hf_xxx Write permission | Set env vars HF_TOKEN

WHY THIS HAPPENS & WHY MULTIPLE INSTALLS NEEDED: Token alone NOT enough - must accept license. Most missed step by juniors.

STEP 9: API Change Check

COMMAND TO RUN:

```
python transcribe_fixed_meta.py
```

EXPECTED OUTPUT (If OK):

```
Loaded pyannote/speaker-diarization-3.1
```

ACTUAL ERROR (If Broken):

```
TypeError: Pipeline.from_pretrained() got unexpected keyword 'token'
```

WHERE TO CHECK: pyannote 3.1.1 used token=, 3.3.2 uses use_auth_token=, latest uses no arg + env var

HOW TO FIX: Use compatibility wrapper trying token, then use_auth_token, then no-arg (see Section 6)

WHY THIS HAPPENS & WHY MULTIPLE INSTALLS NEEDED: API changed 3 times without deprecation. Always write adapter.

STEP 10: Final Verification

COMMAND TO RUN:

```
python transcribe_fixed_meta.py
```

EXPECTED OUTPUT (If OK):

```
Loaded pyannote/speaker-diarization-3.1 + Processing + SUCCESS -> .txt
```

ACTUAL ERROR (If Broken):

```
Any traceback - capture log and check Section 5
```

WHERE TO CHECK: Check transcripts/ folder - should have .txt with [Speaker] [HH:MM:SS]

HOW TO FIX: If SUCCESS, done. If fail, go to Section 5 Workaround.

WHY THIS HAPPENS & WHY MULTIPLE INSTALLS NEEDED: Final gate - includes ffmpeg extraction, Whisper, Pyannote, alignment.

5. Workaround Implementation - Why Multiple Install/Uninstall Cycles Were Required

Context: This section directly addresses the missing explanation in previous versions: why did we perform multiple installations and uninstallations? This was not random troubleshooting, but a systematic discovery process necessitated by pip's inability to resolve major breaking changes across transitive dependencies. Understanding this prevents future engineers from repeating the same trial-and-error cycle.

The original error was pip's dependency resolver does not take into account all packages. When you run pip install pyannote.audio without pinning, pip installs the latest version of every dependency available in 2026: numpy 2.4.6, torch 2.5.1, torchvision 0.20.0, setuptools 80+, huggingface_hub 1.29.0. However, pyannote.audio 3.1.1 was built and tested in 2023 for numpy 1.x, setuptools 69, huggingface_hub 0.34, torch 2.3. Each latest version broke a different part of the system, requiring targeted uninstall and pin to an older compatible version. Each fix revealed the next breaking change, creating the appearance of multiple install/uninstall cycles.

Three Iterations - Detailed Breakdown

Iteration 1: Initial Install (Broken)

Command:

```
pip install pyannote.audio openai-whisper
```

Result: Installed pyannote.audio 3.1.1 + numpy 2.4.6 + torch 2.5.1 + torchvision 0.20.0 + torchaudio 2.5.1 + setuptools 80.2.0 + huggingface_hub 1.29.0 + pyannote-core 6.0.1. All latest versions.

Why Broken (4 failures at once):

```
1. numpy 2.4.6 removed np.NaN but pyannote 3.1.1 uses np.NaN at inference.py:533 -> AttributeError
2. setuptools 80+ removed pkg_resources but lightning_fabric 2.2.5 needs it -> ModuleNotFoundError
3. torch 2.5.1 + torchvision 0.20 mismatch -> circular import: module torchvision has no attribute extension
4. huggingface_hub 1.29.0 breaks transformers which requires <1.0 -> resolver error
```

Lesson: Lesson: Latest is not greatest. Without requirements_locked.txt, pip creates incompatible matrix. This is why we had to uninstall everything and start over. Never trust pip latest on old tutorial code.

Iteration 2: Downgrade Attempt (Partial Fix)

Command:

```
pip install numpy==1.26.4 setuptools==69.0.3 huggingface_hub==0.35.3
```

Result: Fixed np.NaN, pkg_resources, and HF Hub errors. BUT introduced new error: pyannote-core 6.0.1 requires numpy>=2.0 but you have numpy 1.26.4, and pyannote-pipeline 4.0.0 requires pyannote-core>=6.0.0 but you have 5.0.0.

Why Broken: Pyannote has 5 interdependent packages: pyannote-core, pyannote-database, pyannote-pipeline, pyannote-metrics, pyannote-audio. They must move together as a single era. We downgraded numpy to 1.26.4 to fix pyannote.audio 3.1.1, but pyannote-core 6.0.1 was already installed from Iteration 1 and it requires numpy>=2.0. Mixing old numpy with new core fails. This is a classic diamond dependency problem. Solution: decide on ONE era and lock ALL 5 packages to that era. Old era is core 5.0.0, database 5.0.1, pipeline 3.0.1, metrics 3.2.1, audio 3.1.1, numpy 1.26.4. New era is core 6.0.1, database 6.0.1, pipeline 4.0.0, metrics 4.1, audio 3.3.2, numpy 2.4.6. Do not mix eras.

Lesson: Lesson: Pyannote ecosystem must be upgraded as atomic unit. Downgrading one package without downgrading its 4 siblings causes new conflict. This explains second round of uninstall/reinstall.

Iteration 3: Final Locked Era (Working) - Award-Winning Solution
Command Sequence (Exact Order Matters - This Order Prevents Re-breakage):

```
1. pip uninstall -y torch torchvision torchaudio pyannote.audio pyannote-core
pyannote-database pyannote-pipeline pyannote-metrics
2. pip install setuptools==69.0.3 wheel==0.44.0  (MUST BE FIRST - restores
pkg_resources before anything else imports it)
3. pip install torch==2.4.1 torchvision==0.19.1 torchaudio==2.4.1 --index-url
https://download.pytorch.org/whl/cu121  (MUST BE ATOMIC - all 3 together from
same CUDA index)
4. pip install pyannote.audio==3.3.2 numpy==2.4.6 huggingface_hub==0.35.3 hf-
xet==1.1.5 matplotlib openai-whisper pytorch-lightning==2.2.5
torchmetrics==1.4.1
5. hf auth login + Accept licenses on HF Hub
```

Why Works: pyannote.audio 3.3.2 fixed np.NaN to np.nan for NumPy 2.x compatibility (verified in source code line 533 now uses np.nan). Torch trio 2.4.1/0.19.1/2.4.1 are built together with same CUDA 12.1 and same C++ ABI, so _meta_registrations.py finds extension correctly - no circular import. Setuptools 69.0.3 still ships pkg_resources module that lightning_fabric needs. HF Hub 0.35.3 is less than 1.0 and compatible with transformers 4.57.6. Matplotlib satisfies pyannote's runtime import.

Key Insight: Key Insight: Uninstall ALL conflicting packages first, then reinstall as atomic units: torch trio as one atomic unit, pyannote 5 packages as one atomic unit, setuptools separately first. This is why multiple install/uninstall cycles were required - we were discovering incompatible pairs via trial and error. Future prevention is to use requirements_locked.txt with ALL versions pinned and never rely on pip latest. This is the difference between junior and senior approach - senior pins everything on day one.

Complete Reset Procedure - Copy-Paste Final Working

Step 1: Delete old broken venv:

```
Remove-Item -Recurse -Force venv
```

Why: Removes all broken packages. Thirty years experience: never fix broken venv, recreate. Faster and guarantees clean state.

Step 2: Create fresh venv:

```
python -m venv venv
```

Why: Creates isolated environment.

Step 3: Activate venv:

```
.\venv\Scripts\Activate.ps1
```

Why: Prompt shows (venv). All pip installs go to venv.

Step 4: Pin setuptools FIRST:

```
pip install setuptools==69.0.3 wheel==0.44.0
```

Why: 69 is last with pkg_resources. If you install torch first, it pulls setuptools 80+ and breaks. Order matters: setuptools first.

Step 5: Install Torch Trio ATOMICALLY:

```
pip install torch==2.4.1 torchvision==0.19.1 torchaudio==2.4.1 --index-url
https://download.pytorch.org/whl/cu121
```

Why: Must install all 3 together from same index in ONE command. Never separately. They share C++ extensions that must match.

Step 6: Install Pyannote Locked Era:

```
pip install pyannote.audio==3.3.2 huggingface_hub==0.35.3 hf-xet==1.1.5
openai-whisper matplotlib
```

Why: 3.3.2 fixes np.NaN, 0.35.3 fixes HF Hub 1.0 break, matplotlib required even if you don't plot.

Step 7: HF Auth + License:

```
hf auth login + Accept at huggingface.co/pyannote/speaker-diarization-3.1
```

Why: Token alone NOT enough - must accept license. Write permission required.

Step 8: Verify:

```
python -c "from pyannote.audio import Pipeline; print('pyannote ok')"
```

Why: Must print pyannote ok + only harmless UserWarning about torchaudio backend - IGNORE warning.

6. Critical Code Adapters - Copy-Paste Exactly

Context: The previous documentation missed these three adapters, which is why your colleague understood the issue but could not implement the workaround. These adapters are mandatory for the pipeline to work with pyannote.audio 3.3.2. Each adapter fixes a specific breaking API change that occurred between version 3.1.1 and 3.3.2. Without these adapters, even a perfectly pinned environment will fail.

Adapter 1: Fixes DiarizeOutput -> Annotation Type Change

Problem: In pyannote.audio 3.1.1, pipeline(audio_path) directly returned an Annotation object with itertracks() method. In pyannote.audio 3.3.2, it returns a DiarizeOutput wrapper object that does NOT have itertracks(). The actual Annotation is nested inside attributes like speaker_diarization. Code using output.itertracks() fails with AttributeError.

```
def extract_annotation(output):
    if hasattr(output, "itertracks"):
        return output  # Old API 3.1.1 - already Annotation
    for attr in ("speaker_diarization", "diarization", "annotation"):
        if hasattr(output, attr):
            ann = getattr(output, attr)
            if ann and hasattr(ann, "itertracks"):
                return ann
    return output

# Usage:
diarization_output = pipeline(audio_path)
diarization = extract_annotation(diarization_output)
for turn, _, speaker in diarization.itertracks(yield_label=True):
    print(turn, speaker)
```

Adapter 2: Fixes token -> use_auth_token API Change

Problem: Authentication argument changed three times: 3.1.1 used token=, 3.3.2 uses use_auth_token=, latest uses no arg + HF_TOKEN env var + cached login. Code using token= fails on 3.3.2 with TypeError: got unexpected keyword argument 'token'.

```
def load_pipeline():
    import os
    os.environ["HF_TOKEN"] = HUGGINGFACE_TOKEN
    os.environ["HUGGINGFACE_TOKEN"] = HUGGINGFACE_TOKEN
    os.environ["HUGGING_FACE_HUB_TOKEN"] = HUGGINGFACE_TOKEN
    for model_id in ["pyannote/speaker-diarization-3.1"]:
        try:
            try:
                pipeline = Pipeline.from_pretrained(model_id,
token=HUGGINGFACE_TOKEN)  # Old 3.1.1
            except TypeError:
                try:
                    pipeline = Pipeline.from_pretrained(model_id,
```

```
use_auth_token=HUGGINGFACE_TOKEN)  # New 3.3.2
                except TypeError:
                    pipeline = Pipeline.from_pretrained(model_id)  # Cached login
            pipeline.to(device)
            return pipeline
        except Exception as e:
            print(f"Failed {model_id}: {e}")
```

Adapter 3: Fixes Missing Output Directory and Timestamp Parsing

Problem: Two steps missed in previous docs cause FileNotFoundError and incorrect timestamps. Transcripts directory must be created with mkdir(parents=True, exist_ok=True) before writing. Video names in format VID_YYYYMMDD_HHMMSSmmm contain recording timestamp that must be parsed to absolute timestamps like [13:03:35.125]. Without parsing, timestamps are relative and not auditable.

```
from pathlib import Path
OUTPUT_DIR = Path(r"D:\...\Videos\transcripts")
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)  # Must have exist_ok=True

import re
from datetime import datetime
def get_base_timestamp(video_name):
    m = re.match(r'VID_(\d{8})_(\d{9})', video_name)
    if m:
        d,t = m.group(1), m.group(2)
        return datetime(int(d[:4]), int(d[4:6]), int(d[6:8]), int(t[:2]),
int(t[2:4]), int(t[4:6]), int(t[6:9])*1000)
    return None

def clear_gpu():
    import gc, torch
    if torch.cuda.is_available():
        torch.cuda.synchronize()
        torch.cuda.empty_cache()
        gc.collect()
        torch.cuda.empty_cache()
```

7. Final Verification Checklist

Purpose: After completing all fixes, verify environment is production-ready using this checklist. Each check must pass before running full transcription. This checklist is designed for junior engineer with no prior context and includes exact commands and expected outputs with differentiated typography as requested.

- **Pyannote Import:**

```
Command: python -c "from pyannote.audio import Pipeline; print('pyannote ok')"
Expected: pyannote ok + harmless UserWarning about torchaudio backend - IGNORE
```

If fails: If fails, go to Section 4 Steps 2-7

- **pkg_resources:**

```
Command: python -c "import pkg_resources; print('ok')"
Expected: ok
```

If fails: If fails, setuptools 80+ installed - pip install setuptools==69.0.3

- **Torch Trio:**

```
Command: python -c "import torch, torchvision; print(torch.__version__,
torchvision.__version__)"
Expected: 2.4.1 0.19.1 - must match exactly
```

If fails: If mismatch or circular import, uninstall trio and reinstall atomically

- **HF Hub Version:**

```
Command: pip show huggingface_hub
Expected: Version: 0.35.3 (must be <1.0)
```

If fails: If 1.29.0, pip install huggingface_hub==0.35.3

- **Transcripts Output:**

```
Command: Check transcripts/ folder after run
Expected: 2 .txt files >1KB with [Speaker] [HH:MM:SS.mmm - HH:MM:SS.mmm] text,
starting with === EVENT DETAILS === + Start/End Time + Duration IST
```

If fails: If missing, check OUTPUT_DIR.mkdir and permissions